

WEIR - Assignment 3 - Retrieval-Augmented Generation

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Abstract—This report describes a Retrieval-Augmented Generation (RAG) system built on top of the PA2 document retrieval pipeline. The system answers user questions in two modes: with context (retrieval + reranking + generation) and without context (LLM-only). We focus on transparent retrieval, explicit evidence in prompts, and the impact of retrieval quality on generated answers. Evaluation and detailed graphs are added in a later iteration.

I. INTRODUCTION

This assignment extends our PA2 retrieval system with a language model to create a RAG pipeline that can answer questions in two modes: (1) *With Context*, where the model receives retrieved document chunks as explicit evidence, and (2) *Without Context*, where the model answers directly from its parameters. The goal is to analyze how retrieval quality affects answer quality, transparency, and failure modes.

II. IMPLEMENTATION SUMMARY

A. Models

Embedding model. We use BAAI/bge-m3 as the default embedding model with an output dimension of 1024. Additional models are supported (e.g., all-MiniLM-L6-v2 with 384 dimensions and all-mpnet-base-v2 with 768 dimensions), selected by an environment variable.

Reranker. We use a cross-encoder reranker, cross-encoder/mmarco-mMiniLMv2-L12-H384-v1, with alternatives available (e.g., ms-marco-MiniLM-L-12-v2, qnli-distilroberta-base).

LLM. Answer generation is performed by an Ollama-hosted model, ollama_chat/llama3.2:1b (approximately 1B parameters), accessed via the DSPy interface.

B. Document storage and chunking

We reuse the PA2 database schema and chunked documents. Chunks are stored in dimension specific tables (page_segment_vec384, page_segment_vec768, page_segment_vec1024) along with page IDs and model IDs. The chunking strategy is sentence-based as described in the PA2 report.

C. Retrieval and reranking

A user query is embedded using the selected embedding model. We then retrieve the top- k segments from pgvector using a configurable similarity metric (cosine, L2, or L1). The retrieved candidates are reranked with a cross-encoder. The final top- n chunks are formatted into a context block that becomes explicit evidence for the With Context mode.

D. Prompt construction and generation

The system defines two DSPy signatures: SimpleRAG (context + question) and SimpleQuery (question only). The With Context mode feeds retrieved evidence plus the user question to the LLM. The Without Context mode submits only the question. Both answers are generated in parallel and printed alongside the retrieved context for traceability.

E. Tools and libraries

The implementation uses DSPy for model orchestration, sentence-transformers for embeddings and reranking, SQLAlchemy and pgvector for retrieval, and Ollama for local LLM inference.

III. SYSTEM EVALUATION CRITERIA

This section will be completed after running evaluation experiments.

A. Data Processing

We will assess how chunk size and sentence-based segmentation affect retrieval granularity and downstream answer quality. We will report trade-offs between short chunks (higher precision) and longer chunks (more context).

B. Data Retrieval

We will report the top- k retrieval count and analyze relevance of retrieved chunks. The impact of retrieval quality on answer correctness will be summarized with examples.

C. Evaluation

We will provide a table comparing With Context vs Without Context answers for at least nine queries (six good and three poor RAG outcomes), and explicitly describe when retrieval improves answer quality, when it fails, and why.

IV. DISCUSSION

We will analyze typical failure modes, distinguishing retrieval errors (irrelevant or missing evidence) from generation errors (hallucination). We will also discuss how explicit evidence improves explainability and user trust.

V. CONCLUSION

We implemented a transparent RAG pipeline on top of the PA2 retrieval system, supporting both evidence-grounded and LLM-only answering modes. The system exposes retrieved evidence and enables a direct comparison of answer quality across the two modes. Evaluation and visualizations will be added after running the experiments.

REFERENCES